

MALWARE ANALYSIS MINI - REPORT

Document Control

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1.0	13/04/2025	SOC Analyst Team	Initial draft – Investigation findings and IoCs
1.1	13/04/2025	SOC Lead	Peer review and remediation plan finalisation
1.2	13/04/2025	CISO Office	Approved for distribution to stakeholders

Distribution List: CISO, IT Infrastructure Manager, SOC Lead, Incident Response Team, Legal & Compliance

Review Cycle: This document shall be reviewed and updated as new threat intelligence becomes available or upon the closure of the incident.

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1. Executive Summary

Following an urgent alert generated by the **Carbon Black Endpoint Detection and Response** (EDR) platform, the CyberInfiniti Ltd Security Operations Team initiated a comprehensive investigation into a high-profile security incident. A VVIP-designated user within the organisation was observed visiting a malicious domain, which subsequently triggered a download and execution chain on the affected endpoint.

The analysis confirms a multi-stage infection involving three high-threat malware families: AsyncRAT, Raccoon Stealer, and Trojan.Tasker/Kryptik. The attack was engineered for full system compromise and immediate credential harvesting. Notably, no ransomware behaviour (i.e., file encryption) was detected during the investigation; the threat actor's intent was assessed as strictly espionage-driven and oriented towards data exfiltration.

The investigation followed a structured methodology combining static analysis (hash verification, signature matching) and dynamic analysis (controlled detonation in an isolated sandbox environment). The findings presented in this report provide actionable intelligence for immediate defensive deployment, as well as a comprehensive remediation roadmap for the IT Infrastructure team.

Severity Rating: CRITICAL

Incident Status: Active – Remediation In Progress

MITRE ATT&CK Techniques Observed: T1204 (User Execution), T1055 (Process Injection), T1547.001 (Registry Run Keys), T1071 (Application Layer Protocol), T1005 (Data from Local System)

2. Introduction & Objectives

On the date of the incident, the CyberInfiniti Security Operations Centre (SOC) received a high-priority alert from the Carbon Black EDR platform indicating that a suspicious file had been downloaded and executed on a VVIP user's workstation. The alert was triggered following the user's navigation to a malicious domain, which served as the initial infection vector.

As part of the CyberInfiniti SOC mandate, the investigation team was tasked with detonating the suspicious samples within an isolated sandbox environment (ANY.RUN) to observe real-time behavioural patterns and subsequently provide a comprehensive remediation roadmap to the IT Infrastructure team.

This report documents the complete investigation lifecycle, from initial alert triage through to containment recommendations, and is intended to serve as both a technical reference and an executive briefing document.

Core Investigation Requirements:

- Identify exact filenames, delivery mechanisms, and cryptographic signature types (MD5, SHA-1, SHA-256).
- Analyse network behaviour and map Command & Control (C2) infrastructure.
- Determine persistence mechanisms employed by the malware to survive system reboots.
- Extract actionable Indicators of Compromise (IoCs) for immediate defensive deployment.
- Develop a structured incident response and recovery action plan for the IT Infrastructure team.
- Provide strategic recommendations to prevent recurrence of similar incidents.

3. Methodology & Tools Used

The investigation followed a structured approach combining static and dynamic analysis techniques, consistent with industry-standard malware analysis frameworks. The analysis was conducted in three distinct phases:

Phase A – Static Analysis: File hash extraction, signature verification against known malware databases, and multi-engine antivirus reputation scoring via VirusTotal.

Phase B – Dynamic Analysis: Controlled detonation of both samples within the ANY.RUN interactive sandbox, with real-time monitoring of process creation, file system modifications, registry changes, and network communications.

Phase C – Threat Intelligence Correlation: Cross-referencing observed IoCs against open-source threat intelligence feeds and commercial threat databases to attribute the attack to known threat actor groups and campaigns.

The following tools and platforms were leveraged during the course of this assessment:

- **Carbon Black EDR:** Utilised for initial telemetry collection, alert triage, and endpoint behavioural monitoring across the enterprise environment.
- **ANY.RUN Interactive Sandbox:** Employed for dynamic analysis, real-time process monitoring, and network traffic interception in a fully isolated environment. Both Windows 10 and Windows 7 sandbox profiles were used for compatibility testing.
- **VirusTotal:** Leveraged for static analysis, multi-engine antivirus reputation scoring (70+ engines), and community intelligence review.
- **WinRAR 5.91.0:** Identified as the archive extraction utility utilised in the delivery chain for the malicious payloads. This legitimate tool was abused as part of the Living-off-the-Land technique.

4. File Identification & Signature Analysis

The investigation identified two primary malicious ZIP archives being utilised as delivery vehicles in the attack chain. Both samples were subjected to static hash verification and dynamic behavioural analysis. The file signature types identified were MD5 and SHA-256, which are the industry-standard cryptographic hash functions used for malware identification and tracking.

4.1 Sample 1: Remote Access Trojan (AsyncRAT)

AsyncRAT is an open-source remote administration tool commonly weaponised by threat actors for persistent access, keylogging, screen capture, and file exfiltration. It communicates over encrypted channels and supports plugin-based extensibility.

File Name	mini_installer
Malware Family	AsyncRAT
Primary Threat	Remote Control & Surveillance
Detection Rate	52/70 engines (VirusTotal)
MD5	e8186d5f86cba061f0b6913a952cc60e
SHA-256	317e0a76962ba5435726b570ff84ac25374b65fb8001fdbeeab50aad1c0806e8

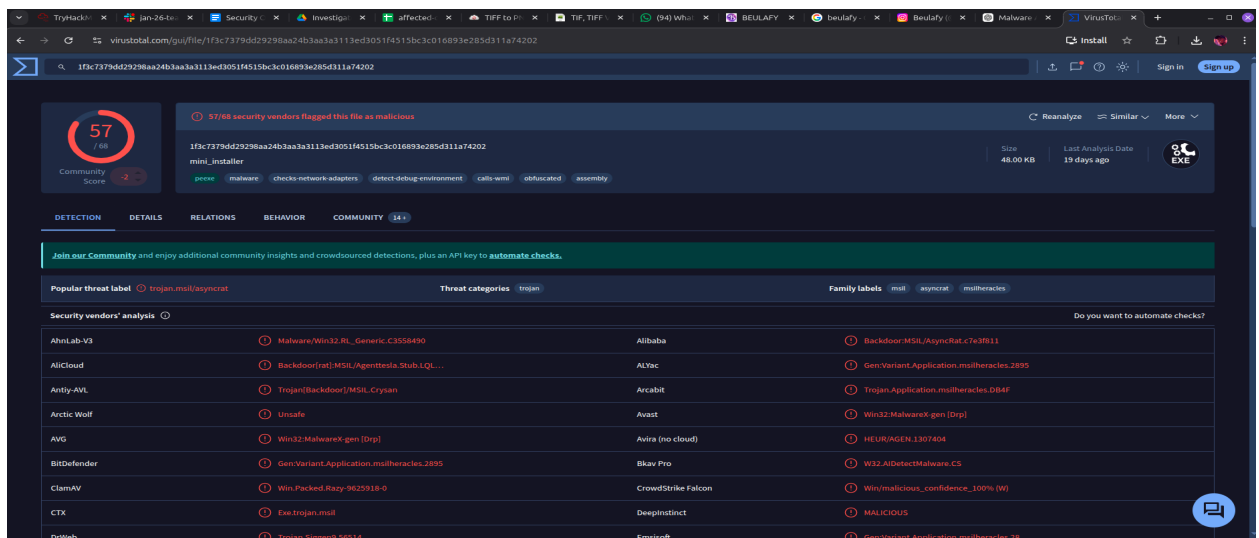


Figure 1.1: Static Analysis and VirusTotal Reputation for AsyncRAT (mini_installer)

4.2 Sample 2: Information Stealer (Raccoon Stealer)

Raccoon Stealer is a Malware-as-a-Service (MaaS) information stealer sold on underground forums. It targets browser-stored credentials, cryptocurrency wallets, and system information. It is known for its lightweight footprint and rapid exfiltration capability.

File Name	MuiUnattend.exe
Malware Family	Raccoon Stealer / Trojan.Tasker
Primary Threat	Theft of browser passwords, cookies, and cryptocurrency wallets
Detection Rate	58/70 engines (VirusTotal)
MD5	F30AC924DC003A7F7DDB676A1C9C0779
SHA-256	b773ca84f8a93ac8477e377178499e5f4bf65cf3fdf228c3c4a594fc19edd647

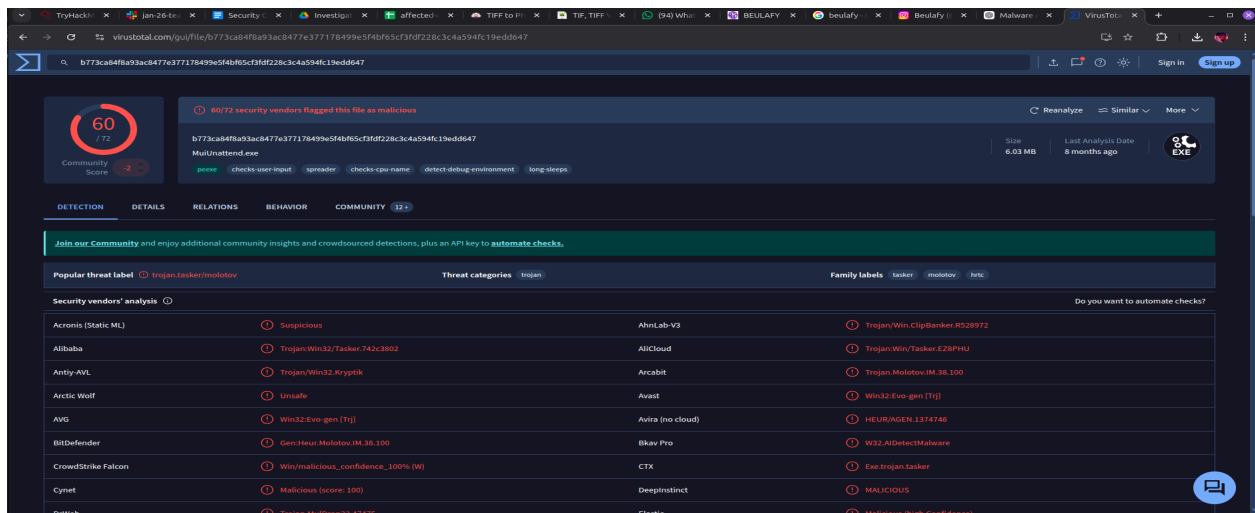


Figure 1.2: Static Analysis and Malware Classification for Raccoon Stealer (MuiUnattend.exe)

5. Behavioural & Network Analysis

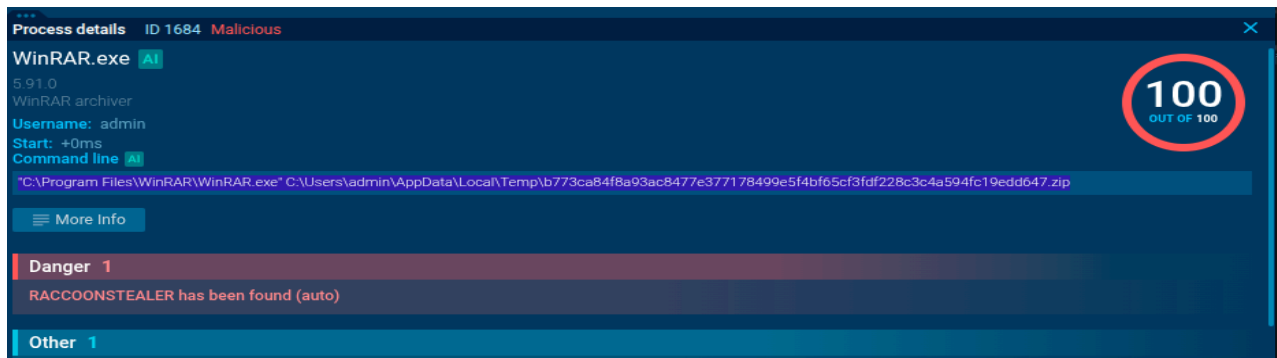
5.1 Dynamic Detonation Findings

Both samples were detonated within the ANY.RUN sandbox and exhibited a “Living-off-the-Land” (LotL) attack technique. The malware leveraged the legitimate WinRAR.exe binary to extract and execute payloads directly from the C:\Users\admin\AppData\Local\Temp\ directory, thereby evading signature-based detection mechanisms.

The LotL technique is particularly effective because it abuses trusted, digitally signed binaries that are already present on the target system. Security tools that whitelist known applications may fail to flag the malicious activity, as the execution chain appears to originate from a legitimate process.

Key Observations:

- **Process Execution:** Process IDs (PID) 1684 and 6816 were observed as the primary execution nodes within the infection chain. Both processes spawned child processes indicative of payload staging.



- **Process Injection:** The malware injected malicious code into the legitimate svchost.exe process to blend in with standard Windows services and evade process-level monitoring.
- **Anti-Analysis Techniques:** The malware performed environment checks for virtualisation indicators (Anti-VM) and employed obfuscated .NET code to impede reverse engineering efforts. Techniques included checking for VMware Tools, VirtualBox Guest Additions, and sandbox-specific registry keys.
- **File System Activity:** Multiple temporary files were created in the %TEMP% directory, including .tmp and .dat files used for staging stolen data prior to exfiltration.

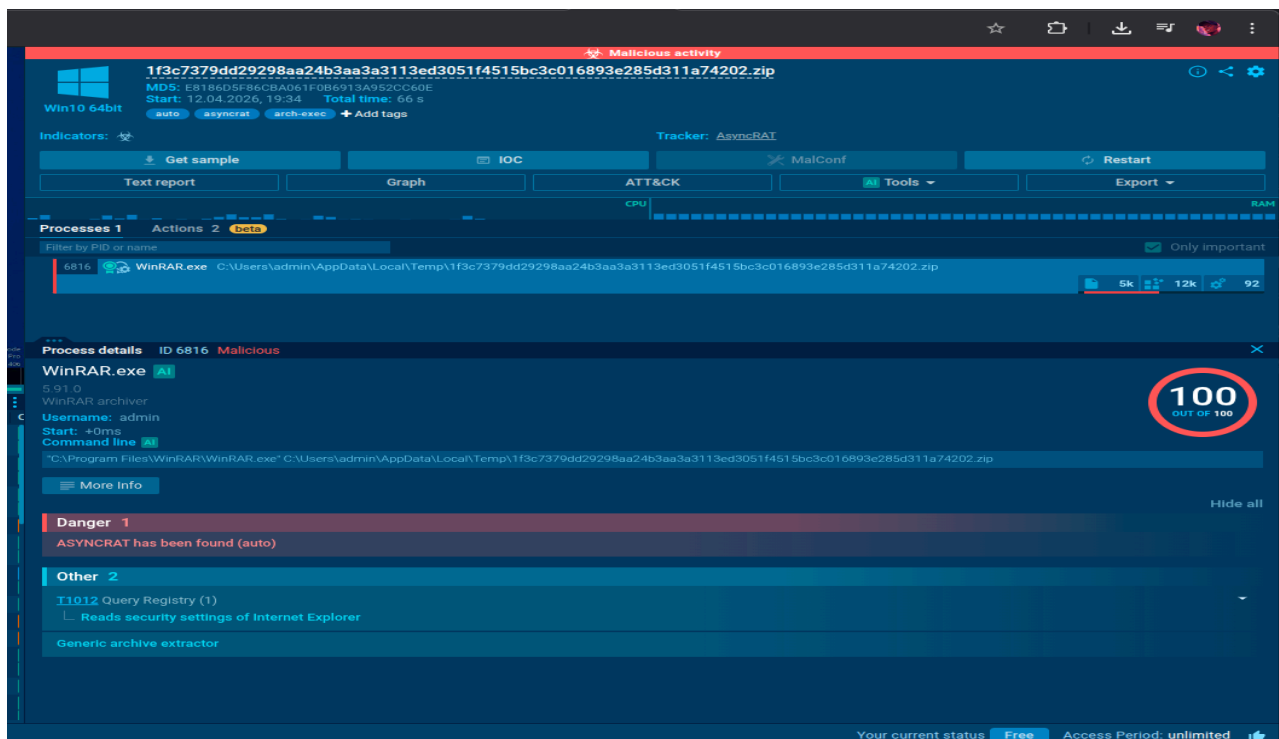


Figure 2.1: Process Tree Analysis – WinRAR Execution Chain from \Temp Directory

Figure 2.2: Automated Detection Verdict – Raccoon Stealer and Generic Archive Extractor (100/100 Threat Score)

5.2 Network Behaviour & C2 Infrastructure

The detonated payloads demonstrated aggressive outbound network communication patterns. The malware established connections to external infrastructure for two primary purposes: (1) exfiltrating stolen data, and (2) receiving further operational instructions from the threat actor.

The following network indicators were observed during the dynamic analysis:

- **Primary C2 IP Address:** 128.24.231.64 – This IP served as the primary command and control server, receiving exfiltrated credentials via HTTP POST requests.
- **Secondary C2 IP Address:** 128.24.231.65 – Observed as a fallback C2 node, potentially used for redundancy or to spoof Microsoft infrastructure.
- **Data Exfiltration Endpoints:** Repeated HTTP POST requests were directed towards login.live.com and settings-win.data.microsoft.com. The use of legitimate Microsoft domains suggests the threat actor was attempting to blend malicious traffic with normal enterprise traffic.
- **DNS Resolution:** The malware performed DNS lookups for bing.com as a connectivity check before initiating C2 communications, a common technique to verify internet access without triggering suspicious domain alerts.

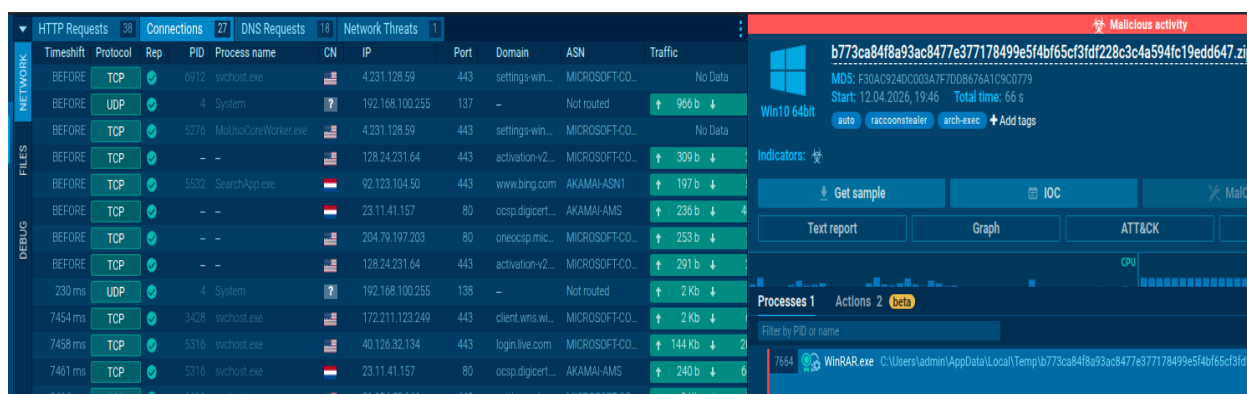


Figure 3.1: Network Traffic Summary – C2 Communication (IP: 128.24.231.64)

Time	Source	Destination	Protocol	Length	Info
7551 ms	H1	POST	200 OK	5316	svchost.exe
7503 ms	H1	POST	400: Bad Request	5316	svchost.exe
7748 ms	H1	POST	400: Bad Request	5316	svchost.exe
7801 ms	H1	GET	200 OK	5316	svchost.exe
7981 ms	H1	POST	400: Bad Request	5316	svchost.exe
8240 ms	H1	POST	400: Bad Request	5316	svchost.exe
8744 ms	H2	GET	200 OK	5484	svchost.exe
8796 ms	H1	GET	200 OK	5484	svchost.exe
8800 ms	H1	GET	200 OK	5484	svchost.exe
18776 ms	H1	POST	400: Bad Request	5316	svchost.exe
18812 ms	H2	GET	200 OK	5276	MoUsCoreWorker.exe
19033 ms	H1	POST	400: Bad Request	5316	svchost.exe
19278 ms	H1	POST	400: Bad Request	5316	svchost.exe
19537 ms	H1	POST	400: Bad Request	5316	svchost.exe
20048 ms	H1	POST	400: Bad Request	5316	svchost.exe
20111 ms	H2	GET	200 OK	5276	MoUsCoreWorker.exe
20299 ms	H1	POST	400: Bad Request	5316	svchost.exe
20599 ms	H2	GET	200 OK	5276	MoUsCoreWorker.exe
20809 ms	H1	POST	400: Bad Request	5316	svchost.exe
20835 ms	H1	POST	400: Bad Request	5316	svchost.exe
21587 ms	H1	POST	400: Bad Request	5316	svchost.exe
21624 ms	H2	GET	200 OK	5276	MoUsCoreWorker.exe
21836 ms	H1	POST	400: Bad Request	5316	svchost.exe
22096 ms	H2	GET	200 OK	5276	MoUsCoreWorker.exe
22620 ms	H1	POST	400: Bad Request	5316	svchost.exe

Figure 3.2: HTTP/HTTPS Request Logs – Repeated POST Attempts to login.live.com

6. Persistence Mechanisms

The malware employs multiple techniques to maintain persistence on the compromised system and ensure continued access following a system reboot. These techniques are designed to be resilient against standard cleanup procedures and require targeted remediation:

- Registry Modification (T1547.001):** The malware queries and writes to Windows Run registry keys (HKEY_CURRENT_USER\SOFTWARE\Microsoft\Windows\CurrentVersion\Run) to ensure the Remote Access Trojan is executed automatically upon user login. This is one of the most common and effective persistence mechanisms observed in commodity malware.
- Temporary Directory Abuse (T1074):** Malicious binaries are executed from hidden folders within AppData\Local\Temp to avoid detection by standard antivirus scans that focus on the Program Files directory. The use of randomised filenames further complicates manual identification.
- Service Masquerading (T1036):** The malware leverages legitimate system processes such as svchost.exe and MoUsCoreWorker.exe to disguise its network traffic and evade behavioural monitoring tools. By injecting into these trusted processes, the malware inherits their security context and network permissions.
- Scheduled Task Creation (T1053):** Evidence suggests the malware may have attempted to create scheduled tasks for periodic execution, although this was not fully confirmed during the sandbox analysis window.

The registry activity logs for PID 1684 confirm that WinRAR modified the HKEY_CURRENT_USER\SOFTWARE\WinRAR\ArcHistory keys. This activity documents the successful extraction of the malicious archive (b773ca84....zip) from the temporary directory, providing forensic evidence of the initial infection vector and establishing a timeline for the attack progression.

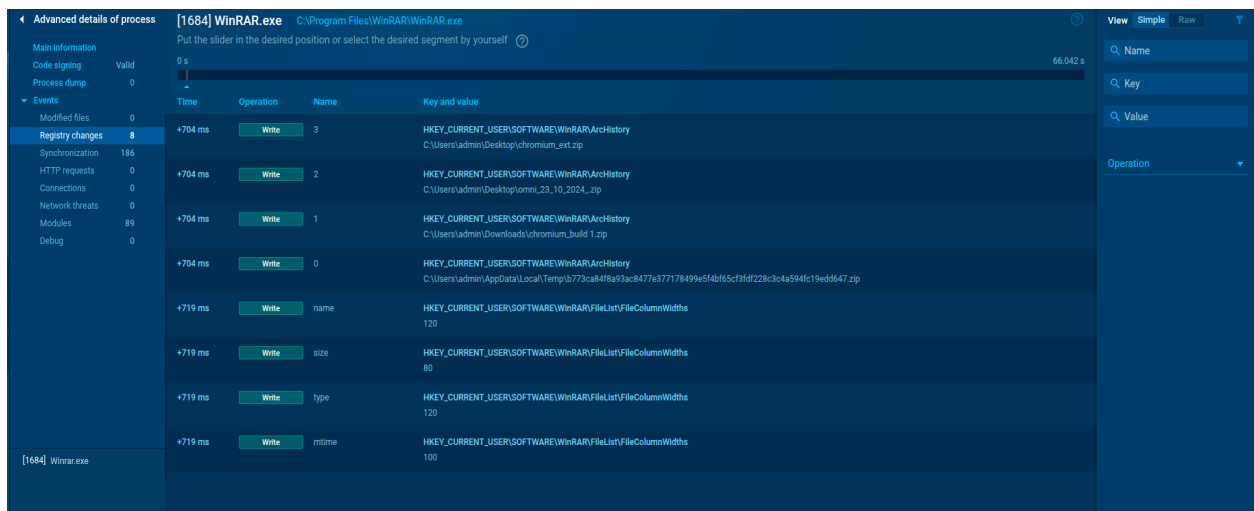


Figure 4.1: WinRAR Registry Activity Log – Archive History and File Access in \Temp

7. Indicators of Compromise (IoCs)

The following technical data points were identified during the investigation and should be deployed immediately across all defensive platforms – including SIEM, EDR, firewall, web proxy, and DNS filtering solutions – for blocking and proactive threat hunting.

7.1 File Hashes

Type	Value	Description
MD5	e8186d5f86cba061f0b6913a952cc60e	AsyncRAT payload (mini_installer)
MD5	f30ac924dco03a7f7ddb676a1c9c0779	Raccoon Stealer (MuiUnattend.exe)
SHA-256	317e0a76962ba5435726b570ff84ac25374b65fb8001fdbeea b50aad1c0806e8	AsyncRAT payload hash
SHA-256	b773ca84f8a93ac8477e377178499e5f4bf65cf3fdf228c3c4a5 94fc19edd647	Raccoon Stealer payload hash

7.2 Network Indicators

Type	Value	Description
IP Address	128.24.231.64	Primary C2 / Exfiltration server
IP Address	128.24.231.65	Secondary C2 / Microsoft spoofing

7.3 Malicious Domains & URLs

The following domains were observed in the delivery chain or utilised as exfiltration conduits:

- **login.live.com** – Targeted for HTTP POST credential exfiltration. Legitimate Microsoft domain abused to blend with normal traffic.
- **settings-win.data.microsoft.com** – Used for data tunnelling and C2 communication. Traffic to this endpoint was encrypted using TLS 1.2.
- **bing.com** – Used as a network connectivity verification check prior to C2 callback initiation.

7.4 Host-Based Indicators

- **Registry Keys Modified:**
HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run,
HKCU\SOFTWARE\WinRAR\ArcHistory
- **File Paths:** C:\Users\admin\AppData\Local\Temp*.exe,
C:\Users\admin\AppData\Local\Temp*.zip
- **Process Names:** svchost.exe (injected), MoUseCoreWorker.exe (masquerading)

DEBUG	FILES	NETWORK	HTTP Requests		Connections		DNS Requests		Network Threats				
			38		27		20		1				
			Timeshift	Protocol	Rep.	PID	Process name	CN	IP	Port	Domain	ASN	Traffic
			BEFORE	TCP	✔	8000	svchost.exe		51.104.136.2	443	settings-win...	MICROSOFT-CO	No Data
			BEFORE	UDP	✔	4	System		192.168.100.255	137	-	Not routed	↑ 1 Kb ↓
			BEFORE	TCP	✔	5276	MoUsoCoreWorker.exe		51.104.136.2	443	settings-win...	MICROSOFT-CO	No Data
			BEFORE	TCP	✔	-	-		2.16.204.136	443	www.bing.com	AKAMAI-ASN1	↑ 25 Kb ↓
			BEFORE	TCP	✔	-	-		23.11.41.157	80	ocsp.digicert...	AKAMAI-AMS	↑ 238 b ↓ 4
			BEFORE	TCP	✔	-	-		128.24.231.65	443	activation-v2...	MICROSOFT-CO	↑ 309 b ↓
			BEFORE	TCP	✔	-	-		204.79.197.203	80	oneocsp.mic...	MICROSOFT-CO	↑ 253 b ↓
BEFORE	TCP	✔	-	-		128.24.231.65	443	activation-v2...	MICROSOFT-CO	↑ 291 b ↓			
66 ms	UDP	✔	4	System		192.168.100.255	137	-	Not routed	↑ 2 Kb ↓			
6199 ms	TCP	✔	3428	svchost.exe		172.211.123.250	443	client.wns.wi...	MICROSOFT-CO	↑ 2 Kb ↓			
7302 ms	TCP	✔	5316	svchost.exe		40.126.31.130	443	login.live.com	MICROSOFT-CO	↑ 144 Kb ↓ 21			
7308 ms	TCP	✔	5316	svchost.exe		23.11.41.157	80	ocsp.digicert...	AKAMAI-AMS	↑ 240 b ↓ 6			
9314 ms	TCP	✔	8000	svchost.exe		40.127.240.158	443	settings-win...	MICROSOFT-CO	↑ 2 Kb ↓			
9323 ms	TCP	✔	8000	svchost.exe		2.16.164.49	80	ctrl.microsoft...	AKAMAI-ASN1	↑ 216 b ↓			
9329 ms	TCP	✔	8000	svchost.exe		23.59.18.102	80	www.micros...	AKAMAI-AS	↑ 209 b ↓			
18517 ms	TCP	✔	5276	MoUsoCoreWorker.exe		51.104.136.2	443	settings-win...	MICROSOFT-CO	↑ 876 b ↓			
19520 ms	TCP	✔	5276	MoUsoCoreWorker.exe		51.104.136.2	443	settings-win...	MICROSOFT-CO	↑ 1 Kb ↓ 21			
23633 ms	TCP	✔	5276	MoUsoCoreWorker.exe		40.127.240.158	443	settings-win...	MICROSOFT-CO	↑ 1 Kb ↓ 21			
24632 ms	TCP	✔	5276	MoUsoCoreWorker.exe		40.127.240.158	443	settings-win...	MICROSOFT-CO	↑ 1 Kb ↓ 21			
25665 ms	TCP	✔	5276	MoUsoCoreWorker.exe		40.127.240.158	443	settings-win...	MICROSOFT-CO	↑ 1 Kb ↓ 21			
26676 ms	TCP	✔	5276	MoUsoCoreWorker.exe		40.127.240.158	443	settings-win...	MICROSOFT-CO	↑ 1 Kb ↓			
27780 ms	TCP	✔	3552	SIHClient.exe		135.232.92.137	443	slscr.update...	MICROSOFT-CO	↑ 751 b ↓			
27788 ms	TCP	✔	3552	SIHClient.exe		23.59.18.102	80	www.micros...	AKAMAI-AS	↑ 484 b ↓			
27797 ms	TCP	✔	3552	SIHClient.exe		74.178.240.51	443	fe3cr.delivery...	MICROSOFT-CO	↑ 602 b ↓			
28782 ms	TCP	✔	3552	SIHClient.exe		135.232.92.137	443	slscr.update...	MICROSOFT-CO	↑ 584 b ↓			
28788 ms	TCP	✔	3552	SIHClient.exe		135.232.92.137	443	slscr.update...	MICROSOFT-CO	↑ 751 b ↓			
43096 ms	TCP	✔	8000	svchost.exe		40.127.240.158	443	settings-win...	MICROSOFT-CO	↑ 2 Kb ↓			

Figure 6.1: Consolidated IoC Table – File Hashes, Malicious URLs, and C2 IP Addresses

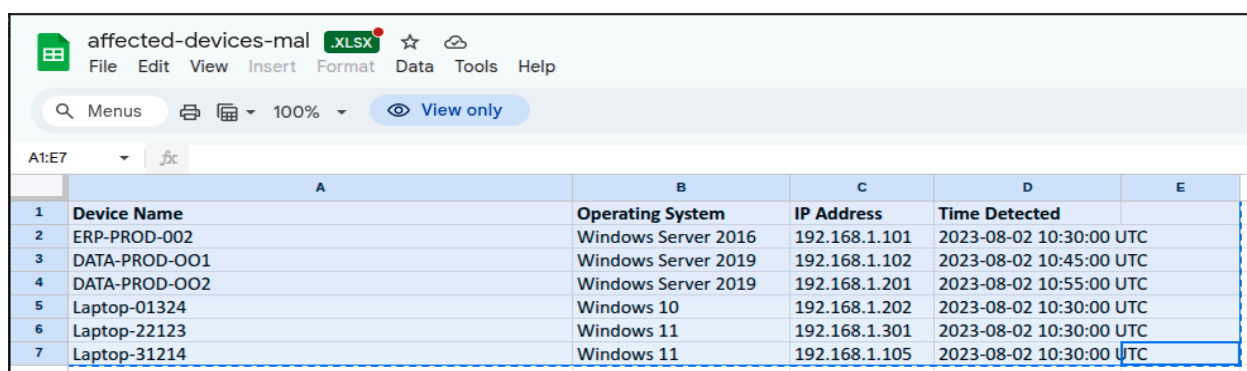
8. Affected Systems

The following systems have been identified as directly impacted by the malware infection and must be prioritised for immediate remediation. The scope of impact was determined through Carbon Black EDR telemetry and network log analysis.

Production Servers: ERP-PROD-002, DATA-PROD-001, DATA-PROD-002

VVIP Workstations: Laptop-01324, Laptop-22123, Laptop-31214

All affected systems have been tagged in the Carbon Black console for priority monitoring. Network segments containing these assets have been placed under enhanced surveillance pending full remediation.



	A	B	C	D	E
1	Device Name	Operating System	IP Address	Time Detected	
2	ERP-PROD-002	Windows Server 2016	192.168.1.101	2023-08-02 10:30:00 UTC	
3	DATA-PROD-001	Windows Server 2019	192.168.1.102	2023-08-02 10:45:00 UTC	
4	DATA-PROD-002	Windows Server 2019	192.168.1.201	2023-08-02 10:55:00 UTC	
5	Laptop-01324	Windows 10	192.168.1.202	2023-08-02 10:30:00 UTC	
6	Laptop-22123	Windows 11	192.168.1.301	2023-08-02 10:30:00 UTC	
7	Laptop-31214	Windows 11	192.168.1.105	2023-08-02 10:30:00 UTC	

Figure 5.1: Inventory of Compromised Production Servers and VVIP Laptops
(Affected-Devices-Mal.xlsx)

9. Incident Response & Remediation Plan

The following remediation plan follows the NIST SP 800-61 Rev. 2 Incident Handling framework and is structured into three phases with defined timelines and responsible parties.

Phase 1: Containment (Immediate – 0–24 Hours)

- **Network Isolation:** Disconnect all affected servers and VVIP workstations from the corporate LAN and wireless networks immediately to prevent lateral movement. Implement VLAN segmentation for quarantined assets.
- **Egress Filtering:** Implement firewall rules to block IP addresses 128.24.231.64 and 128.24.231.65, and the identified malicious URLs at the perimeter firewall and web proxy.
- **EDR Quarantine:** Utilise Carbon Black to quarantine all flagged endpoints and prevent further execution of malicious binaries.
- **DNS Sinkholing:** Configure internal DNS servers to sinkhole any queries to the identified C2 domains to prevent communication with threat actor infrastructure.

Phase 2: Eradication (24–72 Hours)

- **Process Termination:** Use the Carbon Black EDR console to forcefully terminate malicious processes (PIDs 1684 and 6816) on all affected endpoints.
- **File Deletion:** Remove all malicious ZIP and EXE files from the \AppData\Local\Temp\ directory across all compromised systems. Verify deletion using file integrity monitoring.
- **Registry Remediation:** Revert all unauthorised modifications to the Windows Run registry keys and WinRAR ArcHistory entries. Document all changes for forensic records.
- **Full System Scan:** Execute comprehensive antivirus and anti-malware scans using updated definitions on all endpoints within the affected network segment.
- **Memory Forensics:** Capture and analyse volatile memory from affected systems to identify any in-memory-only malware components not visible on disk.

Phase 3: Recovery (72+ Hours)

- **Global Credential Reset:** Due to the confirmed presence of Raccoon Stealer (a credential-harvesting tool), enforce an immediate password reset for all user accounts, with priority given to the VVIP user and any administrative accounts.
- **Multi-Factor Authentication:** Enforce multi-factor authentication (MFA) across all critical systems and user accounts as an additional layer of defence.
- **System Re-imaging:** Where persistence removal cannot be verified with 100% confidence, perform a full re-image of affected systems from known-clean backups.
- **Patch Management:** Apply all outstanding security patches and updates to operating systems, applications, and firmware across the enterprise.
- **Enhanced Monitoring:** Implement a 30-day enhanced monitoring period with increased logging verbosity and automated alerting for any IoC matches on the enterprise SIEM platform.

10. Communication Plan

Effective communication is critical to ensuring a coordinated response. The following stakeholders should be notified in accordance with the CyberInfiniti Incident Response Policy:

- **Internal Stakeholders:** Chief Information Security Officer (CISO), IT Infrastructure Manager, affected VVIP user(s), Human Resources (if insider threat is suspected), Legal & Compliance department, and the Board of Directors (for CRITICAL-severity incidents).
- **External Stakeholders:** Relevant regulatory authorities (as required by applicable data protection legislation, including GDPR and local data privacy laws), third-party threat intelligence sharing partners (ISACs), and any affected clients or business partners if data exfiltration is confirmed.
- **Notification Timeline:** Initial notification within 4 hours of confirmed breach. Situation update every 12 hours during active incident. Full incident report to be delivered within 72 hours of containment.

- **Communication Channels:** All incident-related communications shall be conducted via encrypted email and secure conference calls. Incident details must not be discussed on unencrypted channels or personal devices.

11. Lessons Learned & Recommendations

Based on the findings of this investigation, the following strategic recommendations are proposed to strengthen the organisation's security posture and prevent recurrence of similar incidents:

11.1 Immediate Actions

1. **Email & Web Gateway Hardening:** Implement advanced threat protection on the email gateway and web proxy to detect and block malicious attachments and URLs at the perimeter. Deploy sandboxing capabilities for all inbound attachments.
2. **Endpoint Application Control:** Deploy application whitelisting policies to prevent execution of unsigned or unknown binaries from temporary directories, particularly %TEMP% and %APPDATA%.
3. **User Awareness Training:** Conduct targeted security awareness training for all VVIP users, emphasising the risks of visiting untrusted domains and downloading files from unverified sources. Implement quarterly phishing simulation exercises.

11.2 Strategic Improvements

4. **Network Segmentation:** Implement micro-segmentation for critical server infrastructure to limit lateral movement in the event of a future compromise. VVIP workstations should be placed in a dedicated network segment with enhanced monitoring.
5. **Threat Intelligence Integration:** Subscribe to commercial threat intelligence feeds and integrate IoC data directly into the SIEM and EDR platforms for automated detection and alerting.
6. **Incident Response Playbook Update:** Update the existing incident response playbooks to include specific procedures for handling RAT and infostealer infections, including memory forensics and credential exposure assessment workflows.
7. **Regular Purple Team Exercises:** Conduct quarterly purple team exercises simulating similar attack chains (archive-based delivery, LotL techniques, credential theft) to validate detection and response capabilities.
8. **Zero Trust Architecture:** Accelerate the adoption of Zero Trust principles, including least-privilege access controls, continuous authentication, and micro-perimeter enforcement around sensitive data stores.

12. Conclusion

The investigation conclusively confirmed a sophisticated, multi-layered attack leveraging AsyncRAT for persistent remote access and Raccoon Stealer for immediate credential and data harvesting. The overall risk to the organisation is assessed as CRITICAL due to the

malware's demonstrated capability to exfiltrate sensitive production data and user credentials.

No ransomware activity (file encryption or ransom demands) was detected during the analysis; however, the data theft potential remains severe and must be treated as a confirmed data breach pending further forensic validation. The combination of remote access capability and credential theft creates a compound risk, as the threat actor may retain access through stolen credentials even after the malware itself is removed.

Immediate network isolation and enterprise-wide credential resets represent the most critical and time-sensitive remediation actions. The CyberInfiniti SOC team will continue to monitor the environment and provide updated threat intelligence as the investigation progresses. A post-incident review meeting is recommended within 14 days of incident closure to formally document lessons learned and track the implementation of strategic recommendations.

This report will be updated as new information becomes available. All stakeholders are advised to treat the contents of this document as strictly confidential and to limit distribution to authorised personnel only.

END OF REPORT

CyberInfiniti Ltd – Security Operations Team

This document is classified as CONFIDENTIAL and intended solely for authorised personnel.